INTER IIT TECH MEET 15.0

IIT Guwahati Tech Board

Inter IIT Bootcamp

#GoldforGuwahati

Phase 2

MLPS

Team Size up to 1-3 Members

Submission Deadline 7th October

INTER IIT TECH MEET 15.0

IIT Guwahati Tech Board

1. Problem statement

Build an AI-powered meeting assistant that turns a recorded meeting into an accurate transcript and a usable written record. The system must transcribe the audio, refine domain-specific terminology in the transcript, and generate structured meeting minutes, key decisions, and actionable tasks through a coordinated multimodel pipeline.

The application must present the raw transcript, refined transcript, and generated record in an interactive interface. Its outputs must reflect what was said in the recording; where a task owner or deadline was not stated, the system must leave that detail unspecified rather than invent it.

2. Objectives

• Convert recorded meeting speech into a readable transcript using a speech-to-text model.

• Use a language model to correct transcription errors in domain-specific terms while preserving the meaning of the conversation.

• Use a separate language model to produce structured minutes, decisions, and action items.

• Orchestrate the models into one reliable end-to-end workflow with an interactive interface and downloadable results. Detect the ArUco marker (DICT_4X4_50, 0.24 m) on each clue board and decode the QR code beside it.

3. Requirements

Each board shows an ArUco marker (left) and a QR code (right). The QR code contains one line:

Model type	Required role
Speech-to-text model	Transcribe spoken content from the uploaded meeting recording.
Language model for transcript refinement	Correct likely transcription errors, especially domain-specific terms, using the meeting context.
Language model for meeting documentation	Generate structured minutes, key decisions, and action items from the refined transcript.

The two language-model roles must be implemented as distinct processing stages. Teams may choose the models, implementation language, orchestration approach, and interface technology. The selected models and their roles must be identified in the submission.

• The application must accept an English-language meeting audio file and process it from upload through final output in one run.
• The application must retain both the raw and refined transcripts and produce a structured final record.
• The application must handle unsupported, empty, or unreadable files with a clear error message.
• Outputs for a new recording must be generated by the pipeline; prewritten or hardcoded results do not satisfy the requirement.

4. Functional requirements

Audio input and transcription
Allow the user to upload a meeting recording. Convert its spoken content to a raw transcript and make that transcript visible for review. If the audio cannot be processed, explain the problem in the interface.

Domain-aware transcript refinement
Pass the raw transcript to a language model tasked with correcting plausible recognition errors in technical terms, acronyms, and other domain-specific language. Preserve the speaker's intended meaning, including names, numbers, negation, and commitments. Show the refined transcript separately from the raw transcript so users can compare them.

Meeting minutes and task extraction
Pass the refined transcript to a separate language model. Produce a concise meeting summary, organized minutes, a list of key decisions, and a list of actionable tasks. Each task must state the work to be done and include an owner or deadline only when the recording provides one. Do not present a proposal as an agreed decision or an unstated assignment as a confirmed task.

Workflow and interface
Run transcription, transcript refinement, and meeting-documentation generation in the stated order as one workflow. Provide an interactive interface in which a user can upload audio, start processing, inspect the transcripts and final results, and download the outputs. Show clear processing or failure status.

5. Evaluation:

Submissions will be assessed on a previously unseen English-language meeting recording as well as the submitted demonstration and documentation. The rubric totals 100 points.

Criterion	Points	Evidence assessed
Speech transcription	20	Completeness and accuracy of the raw transcript, including names, terminology, numbers, and negation.
Transcript refinement	20	Correction of genuine terminology errors while preserving the content and intent of the recording.
Minutes and decisions	25	Accurate, concise minutes and correct identification of agreed decisions without invented claims.
Action items	15	Complete and actionable task descriptions, with owners and deadlines reported only when stated.
End-to-end application	15	Distinct model stages, successful processing of a new recording, usable interface, exports, and clear error handling.
Submission quality	5	Clear setup instructions, model identification, and sample outputs that match the demonstration.

Reviewers will compare the generated transcript, minutes, decisions, and tasks against the recording and check that the complete pipeline works on new audio.

6. Output requirements :

The application must display and allow download of the following outputs for each processed recording:

Output	Required contents
Raw transcript	The speech-to-text result before language-model refinement.
Refined transcript	The transcript after domain-aware terminology correction.
Meeting minutes	A concise summary and organized account of the main discussion points.
Key decisions	A structured list of decisions reached in the meeting; an empty list if none were reached.
Action items	A structured list of tasks, with description and any stated owner or deadline; an empty list if none were assigned.

The meeting record must be available in a human-readable format and in a machine-readable structured format. Both formats must convey the same decisions and tasks. Missing owners or deadlines must be shown as unspecified, not guessed.

7. Deliverables

• A source repository or archive containing the complete application, prompts or model instructions, dependency information, and a README with setup and run instructions.

• A working interactive application that can process a new meeting recording from audio upload to downloadable results.

• A short technical description identifying the speech-to-text model and the two language models, explaining each model's role and how outputs move between stages.

• At least one meeting recording that may be shared, together with the raw transcript, refined transcript, minutes, decisions, and action items generated by the application.

• A demonstration showing an end-to-end run and the resulting outputs.